



Fig. 6: Linear resonant actuator (haptic sensor)



Fig. 7: Eccentric rotating mass vibration motor (ERM)

In some enhanced applications, it might be worth to consider an external lens in front of the emitter to create a slim IR curtain only above the screen. This might also be advantageous in terms of power saving issues, as it allows reduction in IRED drive current.

## Haptic sensors

Haptic refers to the sense of touch and is a technology that adds feedback to electronic devices (like tablets and smartphones) through the use of vibration (Fig. 6). The vibration produced by these electronic devices uses two actuators, namely, eccentric rotating mass vibration motors (ERMs) (Fig. 7) and linear resonant motors (LRAs).

In ERMs, vibrations are created in two different perpendicular directions to the rotation axis, while vibrations created by an LRA are based on the movement of a mass that causes repeated displacement in only one direction of axis of linear movement.

ERM has an off-centre load; when it rotates, the centripetal force causes the motor to move. The rotation is created by applying current to armature windings attached to the motor shaft. As these are inside a magnetic field created by permanent magnets on the inside of the motor's body, a force is created, causing the shaft to rotate. To ensure the rotation continues in the same direction, current in the windings is reversed. This is

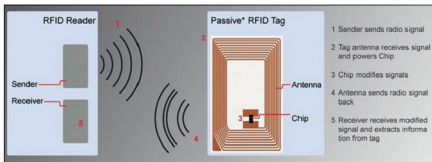


Fig. 8: RFID transceiver

achieved by using static metal brushes at motor terminals, which connect to a commutator that rotates with the shaft and windings. Different segments of the commutator connect to the brushes during rotation and the current is reversed, maintaining the direction of the rotation.

Working of an LRA is similar to how a loudspeaker creates music. In a loudspeaker, the speaker cone is used to produce audio waves through displacement. However, a loudspeaker is designed to operate over a range of frequencies, whereas an LRA is tuned to its resonant frequency. LRAs use magnetic fields and electrical currents to create a force.

One major difference is that the voice coil (equivalent of armature windings) remains stationary and the magnetic mass moves instead. The mass is also attached to a spring, which helps it return to the centre. Driving the magnetic mass up and down causes displacement of the LRA and thereby the vibration force.

## Wireless sensors

Wireless sensors are the standard measurement tools equipped with transmitters to convert signals from process-control instruments into radio transmission. The radio signal is interpreted by a receiver, which then converts the wireless signal to a specific desired output, such as analogue current or data analysis via embedded hardware or computer software.

There are mainly four popular wireless sensors, namely, near field communication (NFC), radio frequency identification (RFID), Bluetooth

and ZigBee.

**Near field communication.** NFC is a form of short-range wireless communication where the antenna used is much smaller than the wavelength of the carrier signal and works on either a magnetic field or an electrical field, but does not support electromagnetic fields for wireless communication.

Normally, in mobile phones' electrical fields, NFC used for short-distance communication operates at 13.56MHz at a wavelength of 22.11m. These are often used in applications like car parking and at ticket counters.

**Radio frequency identification.** This refers to small electronic devices that consist of a small chip and an antenna. This technology uses small I2C memory chips, enough to track items at a distance.

An RF identification system (Fig. 8) uses tags or labels attached to objects to be identified. Two-way radio transmitter-receivers called interrogators or readers send a signal to the tag and read its response. The chip is typically capable of carrying 2000 bytes or less of data.

RFID tags can be passive, active or battery-assisted passive (BAP). An active tag has an onboard battery and periodically transmits its ID signal. A BAP has a small battery onboard and is activated when it is in the presence of an RFID reader. A passive tag is cheaper and smaller because it has no battery.

An RFID device serves the same purpose as a barcode or magnetic strip on the back of a credit card or an ATM card; it provides a unique identifier for that object. And, just as